UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

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| COURSE | DATABASE DESIGN AND DEVELOPMENT |
| COURSE CODE | ICT450 |
| EXAMINATION | NOVEMBER 2026 |
| TIME | 3 HOURS |

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of five (5) questions.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

*This examination paper consists of 6 printed pages.*

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## QUESTION 1 (20 marks)

a) Organizations use data to keep track of their day-to-day operations. Such data is likely to be managed most efficiently when it is stored in a database. Explain the meaning of data and database.

(4 marks)

b) The Database Management System (DBMS) serves as the intermediary between the user and the database. One of the advantages of DBMS is minimized data inconsistency. Briefly explain data inconsistency and illustrate ONE (1) example.

(4 marks)

c) Zinnirah has been appointed as a junior database developer in ZY Stationary company. She is tasked to create a database system to keep track of its customers, orders, and products. Zinnirah has created a database that consists of three tables which are Customers, Orders, and Products. However, the relational operations become very complex and are likely to lead to system efficiency errors and output errors. State the problem that Zinnirah is facing and illustrate the solution (an ERD) to overcome the problem.

(6 marks)

d) Maintenance is one of the important phases in Systems Development Life Cycle (SDLC). Briefly explain THREE (3) types of maintenance in SDLC.

(6 marks)

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## QUESTION 2 (20 marks)

a) Within a larger information system, the database is subject to a life cycle as well. It is known as Database Life Cycle (DBLC). A properly designed database provides you with access to up-to-date and accurate information. It consists of six phases which are database initial study, database design, implementation & loading, testing & evaluation, operation and maintenance & evolution. Briefly explain FOUR (4) activities in the database initial study.

(8 marks)

b) Describe FOUR (4) limitations of mobile database. Give an example for each limitations.

(8 marks)

c) Table 2 consists of data about staff in Qish Boutique. Data type for StaffID, Salary and DepartmentID is a number. Data type for StaffName is a text. Salary is ranged from 3000-10000. Consider the following data.
Based on the data, there are certain rows will certainly fail to be inserted into database. Identify the rows and give a reason according to table property, why the data cannot be inserted.

(4 marks)

Table 2: STAFF

| StaffID | StaffName | Salary | DepartmentID |
|---|---|---|---|
| 3033 | Siti Ziana | 11000 | 10 |
| 3035 | Fazurah | 3000 | 20 |
| 3036 | Haqim Ahmad | 6000 | 20 |
| 3034 | Ahmad Daniel | 150a | 10 |

## QUESTION 3 (20 marks)

Global Sail is a new volunteer organization that registers individuals with various skills to participate in humanitarian missions. It is important for new volunteers to register their skills to ensure appropriate task delegation. The years of experience for each skill are also recorded in the table. Currently, the organization stores all volunteer information in a single spreadsheet, as shown in Table 1. However, this spreadsheet contains several issues. Therefore, the data needs to be normalized before it can be converted into a database system, which will be known as the Global Sail Registration System.

Table 1: Volunteer details

| Item | Register ID | Register Date | Volunteer ID | Volunteer Name | Skills | Years of Experience |
|---|---|---|---|---|---|---|
| 1. | 101 | 28/9/2025 | V15 | Rahmat Zainal | J1 - Journaling | 5 |
| 2. | 101 | 28/9/2025 | V15 | | S2 - Counseling | 2 |
| 3. | 102 | 5/10/2025 | V17 | Nur Fauziah | M1 - Medical Doctor | 25 |
| 4. | 102 | 5/10/2025 | V17 | | C1 - Cooking | 15 |
| 5. | 103 | 30/10/2025 | V20 | Haikal Razali | W1 - Wireman | 10 |
| 6. | 103 | 30/10/2025 | V20 | | S2 - Counseling | 5 |
| 7. | 103 | 30/10/2025 | V20 | | C2 - IT Specialist | 15 |

a) State TWO (2) problems that exist in Table 1. Explain your answer with reasons.

(3 marks)

b) Identify the primary key(s) and partial dependencies and/or transitive dependencies.

(6 marks)

c) Normalize Table 1 from first normal form (1NF) until third normal form (3NF). Show all the steps involved.

(11 marks)

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**QUESTION 4** **(20 marks)**

The relational schema below shows the e-Voting database system for one of the local universities in Malaysia. Each student can vote up to two candidates.

```
STUDENT (StudentID, FirstName, LastName, PhoneNumber, FacultyID*)
CANDIDATE (CandidateID, StudentID*)
VOTING (VotingID, VotingDate, StudID*)
VOTING_DETAILS (VotingID*, CandidateID*, VotingTime)
FACULTY (FacultyID, FacultyName)
```

Write the SQL statements based on the following requirements:

a) Calculate the total number of candidates for each faculty.

(4 marks)

b) Display the details of student who do not vote for any candidate.

(4 marks)

c) Identify the total number of voters for each faculty.

(4 marks)

d) Display the full name of candidates that get more than 500 votes.

(4 marks)

e) Identify the total number of student that vote only one candidate.

(4 marks)

## QUESTION 5 (20 marks)

Clean Environment Sdn. Bhd is a new company located in Jasin, Malacca. This company is planning to develop a solid waste recycling database system for better coordination and reporting on the monitoring, collection, and disposal of waste from each locality. As a database designer, you are tasked to design the database system according to the following requirements:

- Record the details of the collector, such as their collector ID, name, and phone number.
- Among the collectors, several are supervisors. Each collector is managed by one supervisor, and each supervisor manages many collectors.
- Each collector is assigned to several collection duties. Each duty involves only one collector. The date for the collection is recorded in the system.
- Each collection duty session is for one locality and uses one vehicle.
- Locality details, such as locality ID, locality name, and postcode, are recorded in the system.
- Vehicle details, such as the plate number, model, and size, are recorded in the system.
- Each locality contains several houses, and each house is located in one locality. The address and category of each house are recorded in the system.
- Each collection duty session collects various types of solid waste.
- Each piece of solid waste has an ID, name, and category.
- The total weight of the collected solid waste is also recorded in the system.

Based on the given information:

a) Construct a complete Entity Relationship Diagram (ERD) with all of the required tables, attributes, and relationships using the Crow's Foot notation that reflects all of the given business rules.

(16 marks)

b) Identify TWO (2) relevant information that you can get from the ERD you constructed in (a).

(4 marks)

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